

# CSMVessel0000000002 Cruise Ship Cabin Resilience Research

## V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Maritime / Cabins

### Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

### V1.0 vs V2.0 Parameter Changes

| Parameter               | V1.0                      | V2.0                                                                      | Consequence if Not Updated                                          |
|-------------------------|---------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|
| Vibration standard      | ISO 6954:2000             | ISO 6954:2024 (revised limits)                                            | Out-of-date limits; V2.0 aligns with current compliance requirement |
| Aegis Embark transfer   | Not explicitly referenced | Full CSMFAB024-038 product map to cabin applications                      | V1.0 missed major product cross-application opportunity             |
| Schumann field in ship  | Not addressed             | Steel hull eliminates SR; KNbO3-BaTiO3 generators restore 7.83 Hz         | Known passenger health issue unaddressed in V1.0 proposal           |
| Bio-feedback monitoring | Not specified             | PVDF-TrFE mattress topper monitors HR + respiration                       | New passenger wellness IoT product identified                       |
| Window film circadian   | Not in V1.0               | Stellar-Tint (CSMFAB034) applied to portholes; 480nm circadian management | Passenger time-zone management wellness opportunity                 |

### Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.